

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 8: Social Choice and Efficiency

Notes on the Chapter

This chapter has two pieces:

1. It begins with Arrow's (Im)Possibility Theorem: The setting is given, the theorem is stated and proved, and various ways "around" the result (give up on universal domain and have only single-peaked preferences; find some way to make interpersonal comparisons of utility that violate IIA) are discussed.
2. Then, the most obvious and standard way to avoid the unhappy conclusion of Arrow's Theorem is given, namely to set your sights lower: Rather than looking for an aggregate social ordering of social states that is complete and transitive, be satisfied with a partial ordering, the Pareto ordering. Pareto efficiency is defined and, where preferences are expressed numerically and the problem is sufficiently convex, the use of Bergsonian social utility functionals (weighted sums of the individuals' utility) to identify the Pareto efficient set of social states is described and illustrated. For the illustrative example, I use Wilson's theory of syndicates (Wilson, 1968). This second piece closes with remarks about why efficiency may not be all it is typically cracked up to be.

I have found this material relatively straightforward to cover in one class session, although I do not go through the details of the proof of Arrow's Theorem, and I do not go in detail into the application to syndicate theory. (I substitute for syndicate theory Problem 8.11, discussed below.)

Two remarks about this material seem worthy of mention. A number of instructors whom I know cover the Gibbard-Satterthwaite Theorem at the same time as they cover Arrow's Theorem. I see real advantages to that, but I decided that setting up for the G-S Theorem would require too many ideas that properly belong in Volume II. You may disagree with this judgment, in which case you'll need to supplement the chapter.

And the proof that the maximization of all Bergsonian social utility functionals will pick out the entire efficient frontier (for convex problems) is the first application in the book of the Separating-Hyperplane Theorem. Chapters 9 through 11 (and then later

chapters on general equilibrium) become symphonies featuring this mathematical result, and it is probably worthwhile to go through what the theorem says (and doesn't say) and how it is employed (for instance, this application sets the pattern for how we show that a particular separating hyperplane involves a nonnegative normal).

A word on terminology may also be helpful: To try to distinguish between the Arrowian concept of a function mapping H -tuples of preference orders into a preference order and the Bergsonian concept of a function mapping utility imputations into a real number, I use the terms *social preference function* for the former and *social utility function* for the latter.

Notes on the Problems

The problems are, for the most part, quite straightforward. Note that Problem 8.3 asks the students to work with the case of single-peaked preferences; the proposition the student is asked to prove (Proposition 8.5) makes matters relatively simple by assuming that there are an odd number of individuals and that each individual's preferences are anti-symmetric, but this is only relatively simple: It will take a fairly good student to see his or her way through the proof. But the proof is given in the *Guide*; you could ask students to read through the proof and then present a summary of how it works. As mentioned above, I use Problem 8.11 to "teach" the second half of the chapter: The graph in Figure 8.4 is, of course, the Pareto frontier in utility imputation space, gotten by a simple spreadsheet analysis (see details following).

Solutions to Non-Starred Problems

■ 8.1. (a) A simple example of a social preference function that satisfies Assumption 8.1, IIA, and unanimity, is total dictatorship: Choose any individual h^0 , and for any array of preferences $(\succeq_h)_{h \in H}$, let the corresponding social preferences $\succeq$ be (identically) $\succeq_{h^0}$. If each $\succeq_h$ in the array $(\succeq_h)_{h \in H}$ is complete and transitive, then clearly $\succeq \equiv \succeq_{h^0}$ is complete and transitive. This function is well defined for any array of preferences. If $x \succ_h y$ for all h , then this is true for h^0 , so $x \succ y$, proving that this social preference function satisfies unanimity. And if $(\succeq_h)_{h \in H}$ and $(\succeq'_h)_{h \in H}$ agree for each h on how they rank some pair x and y , then they agree in particular for h^0 , but then $x \succeq y$ if and only if $x \succeq_{h^0} y$ if and only if $x \succeq'_{h^0} y$ if and only if $x \succeq' y$, which proves IIA is satisfied.

(b) It is evident that this social preference function satisfies unanimity: if everyone strictly prefers x to y , then individual h_1 does in particular, so society strictly prefers x and y . And the rule for deciding how society feels about x versus y depends only on how individuals feels about x versus y , so IIA is evidently satisfied. The rule is defined for any array of individual preferences, so it is clear that the universal domain assumption holds. So all that really needs to be shown is that social preferences are

complete and transitive. Rather than showing this, I will rely on results from Chapter 1, namely Proposition 1.9(b) and (c), and show that social strict preferences $\succ$ are asymmetric and negatively transitive.

Suppose $x \succ y$. This implies that for some $i = 1, \dots, n$, $x \sim_{h_j} y$ for all $j < i$ and $x \succ_{h_i} y$. And $y \succ x$ implies that for some $i' = 1, \dots, n$, $x \sim_{h_j} y$ for all $j < i'$ and $y \succ_{h_{i'}} x$. Now one of $i < i'$ or $i = i'$ or $i' < i$ must hold. In the first case, we have $y \sim_{h_i} x \succ_{h_i} y$, a contradiction. In the second case, $y \succ_{h_i} x \succ_{h_i} y$, a contradiction. And the third case is handled symmetrically to the first. Thus $x \succ y$ and $y \succ x$ is not possible; $\succ$ is asymmetric.

Suppose $x \succ y$, and z is any third element. We know that for some $i = 1, \dots, n$, $x \sim_{h_j} y$ for all $j < i$ and $x \succ_{h_i} y$. For all $j < i$, either $z \succ_{h_j} x \sim_{h_j} y$ or $x \sim_{h_j} y \succ_{h_j} z$ or $x \sim_{h_j} y \sim_{h_j} z$. Let j' be the smallest index such that one of the first two happens, where j' is set equal to i if the third is the case for all $j < i$. If $j' < i$ and the first case holds, then $z \succ y$. If $j' < i$ and the second case holds, then $x \succ z$. If $j' = i$, then by negative transitivity of $\succ_{h_i}$, one of $z \succ_{h_i} y$ or $x \succ_{h_i} z$ must hold; in the first of these subcases, $z \succ y$ while in the second, $x \succ z$. This shows that $z \succ y$ or $x \succ z$, and $\succ$ is negatively transitive.

(c) No, it doesn't exhaust all possibilities. Suppose $n = 1$; that is, there is one individual. Let $\succeq^*$ be any arbitrary (complete and transitive) ordering of X and consider the social preference function defined by $x \succ y$ if $x \succ_h y$ or if $x \sim_h y$ and $x \succ^* y$. That is, we have a sequential dictatorship, where the lone individual is the first dictator, and the arbitrary ordering $\succeq^*$ breaks ties when h is indifferent. Part b implies that this satisfies Assumption 8.1, unanimity, and IIA, but as long as $\succeq^*$ expresses strict preference for some pair, this is not a sequential dictatorship in the terms of b, which (with a single individual) would force indifference when the single individual is indifferent.

And suppose $n > 1$. Go back to part a, where we have an individual h^0 who is a total dictator; i.e., this individual enforces indifference as well as strict preference. This is not a sequential dictatorship, since the other individuals could strictly prefer x to y when h^0 is indifferent, and in a sequential dictatorship, this would mean $x \succ y$, not $x \sim y$.

■ 8.2. Suppose X has two elements, call them x and y . Let social preferences be defined by $x \succeq y$ if half or more of the individuals have $x \succeq_h y$, and $y \succeq x$ if half or more have $y \succeq_h x$. This clearly is well defined for all arrays of individual preferences, it clearly satisfies unanimity (once we show that it gives a preference relation), and it clearly satisfies IIA. But does it give a complete and transitive ordering on $\{x, y\}$? It is complete, because for each individual, either $x \succeq_h y$ or $y \succeq_h x$ or both, half or more of the individuals have to fit into one camp or the other. And it is transitive: Suppose $z \succeq z' \succeq z''$, where each of z, z' , and z'' is either x or y . If $z = z'$ or $z' = z''$, then $z \succeq z''$ follows trivially. So the case of interest is where $z \neq z'$ and $z' \neq z''$. But since there are only two social states, this means $z = z''$, and $\succeq$ is clearly reflexive (that is,

$z \succeq z$ for all z), since a majority (in fact, every) of the h have $z \succeq_h z$ for both z .

Note that for this social preference function, if the number of individuals is even and half have $x \sim_h y$ while the other half have $y \succ_h x$, then social preferences as defined here give $x \sim y$. This doesn't seem entirely sensible, so: In the case of two social states, is this the only social preference function that satisfies Assumptions 8.1, unanimity, and IIA? The answer is, No. Can you produce some others?

■ 8.4. First we show that $x \gg y$ implies $x \succ y$: If $x \gg y$, then $x \succ_h y$ for all h . Therefore, $x \succeq_h y$ for some h , and $x \succ_h y$ for at least one (in fact, all) h , so $x \succ y$. (Well, to be very precise, this is true as long as there is at least one individual $h \in H$. You are likely to have missed it, but right at the start of the chapter the assumption that H is nonempty was slipped in.)

Now suppose $x \succ y$. Then $x \succeq_h y$ for all h and, for some particular h^0 , $x \succ_{h^0} y$. But then $y \not\succeq_{h^0} x$, so $y \not\succ x$. The binary relation $\succ$ is asymmetric. And if $x \gg y$, then (we've shown) $x \succ y$, so it is not the case that $y \succ x$, hence is it not the case that $y \gg x$ (apply the contrapositive of $x \gg y$ implies $x \succ y$). The relation $\gg$ is asymmetric.

Suppose $x \succ y$ and $y \succ z$. Then $x \succeq_h y$ and $y \succeq_h z$ for all h , and by the transitivity of each $\succeq_h$, $x \succeq_h z$ for each h . Moreover, $x \succ_{h^0} y$ for some one h^0 , and $x \succ_{h^0} y \succeq_{h^0} z$ implies $x \succ_{h^0} z$ (go back to Chapter 1 if you aren't sure about this), hence $x \succ z$, and $\succ$ is transitive.

And if $x \gg y$ and $y \gg z$, then $x \succ_h y \succ_h z$ for all h , hence $x \succ_h z$ (since each $\succ_h$ is transitive), therefore $x \gg z$, and $\gg$ is transitive.

Although the Proposition doesn't state this result, it is also true that if $x \succ y \gg z$, or if $x \gg y \succ z$, then $x \gg z$. If this isn't obvious, give the proof.

■ 8.8. (a) The argument given in the text is nearly a proof, but to be one step more formal: Suppose $\alpha_{h^0} = 0$ and (x_{hs}) is a solution to the maximization problem. Since not all the weights are zero, $\alpha_{h'} > 0$ for some h' . Pick any state s^0 , and consider the sharing rule (x'_{hs}) which is identical to (x_{hs}) except that $x'_{h's^0} = x_{h's^0} + 1$ and $x'_{h^0s^0} = x_{h^0s^0} - 1$. Since $U^{h'}$ is strictly increasing and $\alpha_{h'} > 0$, this increases the term in the maximization problem $\alpha_{h'} \pi_{h'}(s^0) U_{h'}(x'_{h's^0})$ but has no impact on the corresponding term for h^0 , since $\alpha_{h^0} = 0$. Hence, this increases the overall objective function value, contradicting the supposed optimality of the original sharing rule.

(b) Suppose the shares are constrained to be non-negative, (x_{hs}) solves the maximization problem for weights (α_h) , and (x'_{hs}) is Pareto superior to (x_{hs}) . Of course, if (x'_{hs}) is Pareto superior to (x_{hs}) , it gives at least as much objective-function value as does (x_{hs}) , which means it gives just as much, since (by assumption) (x_{hs}) is optimal. It is obvious from the proof of Proposition 8.9 that if $\alpha_h > 0$, then the utility imputation to h from (x'_{hs}) can be no larger than the utility imputation from (x_{hs}) (otherwise, (x'_{hs}) would give a larger value of the objective function in the maximization problem, contradicting the supposed optimality of (x_{hs})). Therefore, there must be some h^0 such

that $\alpha_{h^0} = 0$ and h^0 's expected utility under (x'_{hs}) is larger than her expected utility under (x_{hs}) . But since U_{h^0} is strictly increasing and $x_{h^0s} \geq 0$ for all s , this implies that $x'_{h^0s^0} > 0$ for some state s^0 . Consider a third sharing rule (x''_{hs}) which is identical to (x'_{hs}) except in state s^0 ; in that state, set $x''_{h^0s^0} = 0$ and increase the share of everyone else by $x'_{h^0s^0}/(n-1)$, where n is the cardinality of H . This decreases the expected utility for h^0 , but strictly increases the utility of everyone else, and so would increase the value of the objective function (h^0 has zero weight, while someone else has positive weight), which therefore contradicts the supposed optimality of (x_{hs}) .

(c) I'll only describe verbally what is involved: Suppose (x'_{hs}) is Pareto superior to (x_{hs}) . Then for some h^0 , (x'_{hs}) gives higher expected utility than does (x_{hs}) . Using the continuity of U_{h^0} , we can take a bit away from h^0 in some state and still preserve the strict inequality. (If nonnegativity constraints are imposed on the sharing rules, argue that for x' to give higher expected utility than x , x' must give a strictly positive amount to h^0 in some state, and take the bit away in that state.) Divide this bit amongst the other individuals equally, and their expected utility will all (strictly) increase.

The connection between parts (b) and (c) is the idea that, in this context, we know we can increase any individual's utility by an increase, no matter how small (as long as it is strictly positive) in what that individual gets in any state (since all the probabilities are strictly positive). That idea is employed in both parts.

■ 8.9. Suppose (x_{hs}) solves the maximization problem for given weights (α_h) , and individuals h^1 and h^2 , both risk neutral, differ on their assessments of some state s^1 . Assume in particular that $\pi_{h^1}(s^0) > \pi_{h^2}(s^0)$. Let $\pi = \pi_{h^1}(s^0) + \pi_{h^2}(s^0)$, and let δ_s be an S vector defined by $\delta_{s^1} = 1$ and $\delta_s = -\pi/(1-\pi)$ for all $s \neq s^1$. Consider the alternative sharing rule (x'_{hs}) defined by

$$x'_{hs} = \begin{cases} x_{hs}, & \text{for } h \neq h^1, h^2, \\ x_{hs} + \delta_s, & \text{for } h = h^1, \text{ and} \\ x_{hs} - \delta_s, & \text{for } h = h^2. \end{cases}$$

For individual h^1 , if her utility function is $ax + b$ for $a > 0$ (she is risk neutral, remember), her utility is changed by

$$a \sum_s \pi_{h^1}(s) \delta_s = a \left[\pi_{h^1}(s^0) - (1 - \pi_{h^1}(s^0)) \frac{\pi}{1 - \pi} \right].$$

This is strictly positive if $\pi_{h^1}(s^0) > \pi$, which (by design) it is. Meanwhile and by a similar calculation, this also increases the expected utility of h^2 . (Go through the detailed calculation if you don't see it.) So as long as either α_{h^1} or α_{h^2} is strictly positive, this increases the value of the objective function, contradicting the supposed optimality of (x_{hs}) .

As for the second part of this problem, I would like to apply a similar variation to a supposedly optimal sharing rule (x_{hs}) and conclude that it raises the expected utilities of both parties. But in evaluating the change in expected utilities, if their utility functions aren't linear, I need to approximate the changes. Letting $\bar{u}'_1 = \lim_{x \rightarrow \infty} U'_{h^1}(x)$ (using a right-hand derivative where needed) and letting $\underline{u}'_1 = \lim_{x \rightarrow -\infty} U'_{h^1}(x)$ (using left-hand derivatives), we know that $\bar{u}'_1 < \underline{u}'_1$, but everything will work (in terms of individual h^1) as long as

$$\bar{u}'_1 \pi_{h^1}(s^0) - \underline{u}'_1 (1 - \pi_{h^1}(s^0)) \frac{\pi}{1 - \pi} > 0, \quad \text{or} \quad \frac{\bar{u}'_1}{\underline{u}'_1} \frac{\pi_{h^1}(s^0)}{1 - \pi_{h^1}(s^0)} > \frac{\pi}{1 - \pi}.$$

So, what we need, is a bound on how much the derivative of U_{h^1} declines over its full range (and similarly for U_{h^2} , relative to extent to which h^1 and h^2 disagree over the odds of state s^0 . (I'll leave it to you to fill in the details from here.)

■ 8.11. The picture in the hint is the Pareto frontier (in terms of utility imputations) for this problem. It was generated in Excel, using Proposition 8.10. The space of social states is the space $\{x = (x_1^1, x_2^1, x_1^2, x_2^2) \in R_+^4 : x_1^1 + x_1^2 \leq 15, x_2^1 + x_2^2 \leq 20\}$ which is certainly a convex set, and the utility functions are both concave, so the second half of the proposition tells us that the first half applies. And then the first half tells us that every Pareto-efficient point can be obtained by maximizing (over the set of social states) weighted sums of the two utilities. That is, every efficient point is the solution of

$$\text{Maximize } \alpha[6 + 0.4 \ln(x_1^1) + 0.6 \ln(x_2^1)] + (1 - \alpha)[8 + \ln(x_1^2) + \ln(x_2^2)]$$

over all the feasible divisions of the two goods, for some $\alpha \in [0, 1]$. For the two end-points (in terms of α), it is clear that the optimizing division of the resources gives all of both goods to either the first consumer (if $\alpha = 1$) or the second (if $\alpha = 0$), assuming (as is conventional) that $0 \times -\infty = 0$ in such calculations. As for α in the interior (in $(0, 1)$), we can equivalently (to save on algebra) maximize

$$\beta[6 + 0.4 \ln(x_1^1) + 0.6 \ln(x_2^1)] + [8 + \ln(x_1^2) + \ln(x_2^2)],$$

letting β run over $(0, \infty)$. We know, moreover, since both utility functions are strictly increasing, that all the goods will be used up, so we are maximizing

$$\beta[6 + 0.4 \ln(x_1) + 0.6 \ln(x_2)] + [8 + \ln(15 - x_1) + \ln(20 - x_2)],$$

for $0 \leq x_1 \leq 15$ and $0 \leq x_2 \leq 20$. It is obvious that the maximizing values (for any finite β) will be interior, so this can be solved as an unconstrained maximization problem in two variables, with first-order conditions

$$\frac{0.4\beta}{x_1} = \frac{1}{15 - x_1} \quad \text{and} \quad \frac{0.6\beta}{x_2} = \frac{1}{20 - x_2}, \quad \text{or}$$

$$15 = \left(1 + \frac{1}{0.4\beta}\right)x_1 \quad \text{and} \quad 20 = \left(1 + \frac{1}{0.6\beta}\right)x_2.$$

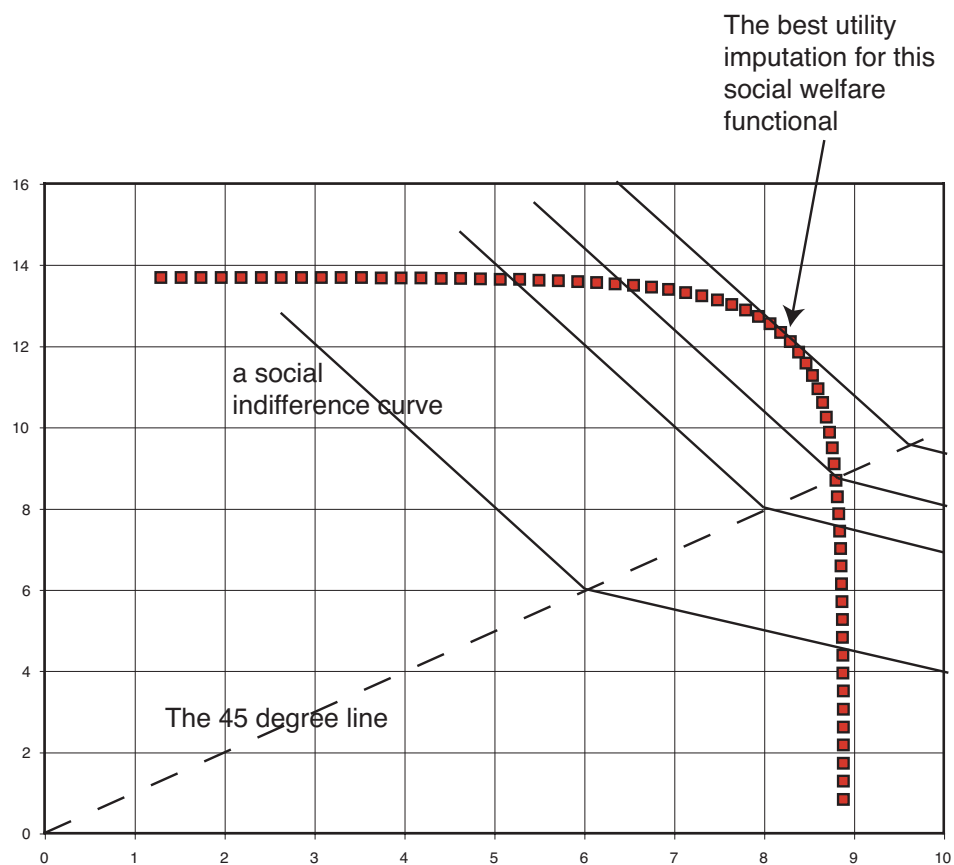
Carrying the algebra one final step, this is

$$x_1 = \frac{6}{0.4 + 1/\beta} \quad \text{and} \quad x_2 = \frac{12}{0.6 + 1/\beta}.$$

Now we can set up a spreadsheet. In column 1 (say), put various values of β . Since I am looking for values of β from 0 to ∞ , I chose $\beta = 0.8^k$, for $k = -31, -30, \dots, 30, 31$. Next, use the formulas to find $x_1, x_2, 15 - x_1$, and $20 - x_2$. Then compute the utilities of the two consumers. With this spreadsheet, I did an X-Y plot of the two utilities, giving the figure.

Now to answer the problem. The social utility functional puts weight 2 on the smaller of the two utilities, and weight 1 on the larger of the two. In the space of utility imputations, we can draw *social indifference curves*, utility imputation pairs that give the same value to the social utility functional. Take, for instance, the point (8,8). This has value 24 according to the social utility functional. If we (say) decrease person 1's utility by 2 units, to 6, to compensate we must increase person 2's utility by 4 units. So the points (6,12) and (12,6) both lie along the social indifference curve that runs through (and has a kink at) (8,8). In the figure shown, I superimpose on the "hint" social indifference curves, which are all parallel translates (with the kink lying along the 45 degree line) of the one we constructed. It is clear from the picture that the optimal social state in terms of imputations will put double the weight on person 1's utility as on person 2's, which corresponds to $\beta = 2$. So I go back to my spreadsheet and look for the solution where $\beta = 2$; it tells me $x_1^1 = 6.667$, $x_2^1 = 10.90909$, $x_1^2 = 8.333$, $x_2^2 = 9.09091$, $u_1 = 8.19$, and $u_2 = 12.327$.

The picture appears on the next page



The Solution to Problem 8.11 in a Picture.